The Zeta Zoo
— The Mathematical Side of Functional Stability Theory

Branch-Local Hilbert-Pólya via the Trinity of UCU, SGE, and Weil-Form Transversality

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Abstract

Functional Stability Theory (FST) is a programme that identifies a single structural pattern—functional positivity under a gauge constraint—as the common substrate of open problems in number theory, mathematical physics, and cosmology [3]. **This paper is the mathematical side of FST.** We formalise the trinity of meta-principles that governs the zeta-type branch of the theory: the *Universal Convexity Uniqueness* (UCU) lemma, which turns strict convexity of the governing functional into the unique saturation state; the *Semigroup-Group Equivalence* (SGE) conjecture, which makes the Hilbert-Pólya property a binary algebraic invariant of the transfer algebra; and the branch-transversal *Weil quadratic form*, which supplies gap positivity across both sides of the SGE dichotomy. Within this trinity, the classical picture reorganises cleanly: the non-existence results NE-A and NE-B of the Riemann v2.1 programme [6] exclude the algebraic routes for $\zeta(s)$; Selberg's Laplace-Beltrami operator realises the Hilbert-Pólya structure explicitly for $Z_\Gamma(s)$; the Spectral Zookeeper [14] closes the Riemann $C^{(2)}$/MS2 step in the CCM/Fourier branch; the conjectural Prime-Hub graph remains open as an explicit OP5 construction under three documented structural obstructions. We classify the *zeta zoo* along the SGE-dichotomy, verify the framework on five test families (Riemann, Selberg, Prime-Hub, Dedekind $\mathbb{Q}(\sqrt{-5})$, Ihara-Petersen), and derive consequences for Dirichlet L-functions [9, 8]. We close with a cryptographic reading of the trinity as the algebraic architecture of the Bitcoin blockchain protocol, and with the narrative circle from the Cooperative Renormalization Model (CRM) of dark energy through FST to the abstract trinity and back via hyperbolic geometry. The physical instantiation of the FST programme, via the Renormalized Free Energy Principle (RFEP), is developed in the companion paper [3].

Contents

1 Introduction

1.1 The Hilbert-Pólya conjecture and its implicit branch choice

The Hilbert-Pólya conjecture asks whether the non-trivial zeros of the Riemann zeta function correspond to the eigenvalues of a self-adjoint operator. Its earliest documented origin is a 1982 letter from G. Pólya to A. Odlyzko [19], recounting a Göttingen conversation with E. Landau around 1912–1914; the first published mention appears only in Montgomery's 1973 pair-correlation paper [20], and the conjecture has been taken up repeatedly since [21, 22, 23]. It has driven much of the analytic number theory of the last century. Yet, as Connes' 2026 survey [24] emphasises, no such operator has been explicitly constructed for $\zeta(s)$.

The programme implicitly makes a branch choice. For the Selberg zeta function $Z_\Gamma(s)$ of a compact hyperbolic surface $\Gamma\backslash\mathbb{H}$, the Laplace-Beltrami operator $\Delta_{\Gamma\backslash\mathbb{H}}$ realises the Hilbert-Pólya structure explicitly: its eigenvalues $\lambda_n = 1/4 + r_n^2$ give $r_n = \mathrm{Im}(\rho_n)$ for the Selberg zeros $\rho_n = 1/2 + ir_n$, and it commutes with all geodesic transfer operators [40]. For the Riemann zeta function, by contrast, Results NE-A and NE-B from the recent Even Dominance proof [6] rule out the two most natural algebraic candidates: the prime-shift Fourier multiplier $M(\xi) = -2\,\mathrm{Re}\,\zeta'/\zeta(\frac{1}{2} + i\xi)$ fails to be absolutely totally positive on a set of positive measure (NE-A), and no non-scalar symmetric matrix simultaneously commutes with all prime-shift matrices at Galerkin truncation $N \leq 15$ (NE-B, with full-space extension conjectural [6]). And for the conjectural Prime-Hub graph (Open Problem 5 of [7]), three obstructions—Weyl-law incompatibility with the logarithmic density $N(T) \sim (T/2\pi)\log(T/2\pi)$, trace-class regularity of a graph transfer operator, and compatibility with NE-A—keep an explicit Prime-Hub/Hilbert-Pólya graph construction open. After the Spectral Zookeeper [14], this openness is no longer the status of the Riemann $C^{(2)}$/MS2 step itself; it is the status of the direct OP5 graph interpretation.

1.2 The trinity of meta-principles

These three canonical facts jointly suggest that the Hilbert-Pólya property is not a universal feature of zeta-type families, but a *local* invariant, present in some sub-families and structurally absent in others. The central organising principle of this paper is that this branch-locality is governed by a trinity of meta-principles that each RH-type programme must address:

(UCU) *Universal Convexity Uniqueness.* The admissible states of the programme are the minimisers of a strictly convex, coercive functional. Strict convexity guarantees a *unique* saturation state, which the spectral machinery of the Weil quadratic form then realises. This principle organises the uniqueness arguments across four otherwise unrelated programmes (turbulence K41, FST dark-energy potential, Riemann Even Dominance, Yang–Mills Gibbs measure) under a single head.

(SGE) *Semigroup-Group Equivalence.* Whether a natural self-adjoint operator—the sought-for Hilbert-Pólya Hamiltonian H_X—exists is conjecturally governed by a single algebraic invariant: whether the transfer algebra $(I_X, \cdot)$ of the family is a (locally compact) group or merely a semigroup. The presence of inverses is precisely the condition under which a Lie structure and hence a Casimir operator emerges.

(WF) *Weil-Form Branch-Transversality.* The Weil quadratic form Q_X is well-defined for every zeta-type family with an Euler product, a gamma factor, and a functional equation, independently of the HP-BL-status. It provides a uniform architecture that yields gap positivity in all algebraic regimes. In the Riemann case, the Q_ζ-based Direct Frontier Dominance of [6] establishes the Riemann hypothesis *unconditionally*, without any Hilbert-Pólya input.

In short:

> *UCU selects the unique saturation state; SGE decides whether a natural Hilbert-Pólya operator exists; Weil-form transversality builds the bridge across the SGE dichotomy.*

The remainder of this paper formalises this trinity, tests it on five zeta-type families, and closes with a speculative reading in which the three principles mirror the algebraic architecture of the Bitcoin blockchain protocol (Section 12).

1.3 A field guide to the zeta zoo

Throughout this paper we use the *zeta zoo* as a pedagogical frame. Each zeta-type family X is an enclosure in the zoo; the enclosure types are the three possible HP-status values (YES, NO, OPEN); the zoo-wide feed is the Weil quadratic form Q_X, which every enclosure accepts regardless of type; the zoo guide is the trinity UCU + SGE + Weil-form transversality, and the four-class rigor hierarchy (Section 13.2). Model enclosures (Riemann, Selberg, Prime-Hub) are the main tour stops. Recent additions (Section 11.2: Dedekind $\mathbb{Q}(\sqrt{-5})$; Section 11.3: Ihara Petersen) sit as first non-canonical zoo exhibits. Species that also inhabit the adjacent FST-physical biotope [3]—*bridge species* such as the Riemann zeta itself, the BSD L-function, the Hodge motive, and the Boolean algebra of P vs NP—are cross-referenced at the relevant entries.

The zoo metaphor is a reading aid, not a theorem scaffold. All mathematical assertions below are formulated technically; the metaphor is used solely to orient the reader through the variety of zeta-type families that the trinity organises.

1.4 The CRM circle: from cosmos to algebra and back

Historically, the Cooperative Renormalization Model (CRM) of cosmological dark energy was the first instantiation of what later became Functional Stability Theory [3]. Its composition law

$$x \oplus y = \frac{x+y}{1+xy}$$

is the Lorentz-boost addition, so the underlying geometry is hyperbolic by construction. When the CRM stability argument was generalised, FST emerged; when FST was abstracted further, the mathematical trinity of UCU, SGE, and Weil-form transversality crystallised. The circle closes: from cosmos to physics to principle to algebra, and the hyperbolic geometry that CRM realises concretely reappears on the YES-side of the SGE dichotomy as the $\mathrm{PSL}_2(\mathbb{R})$-Casimir of Selberg zeta (Section 5) and, on both sides, as the branch-transversal Weil-form architecture (Section 10). At the end, everything is algebra—including the geometry.

1.5 Organization

Section 2 recalls the definitions of the Hilbert-Pólya structure and the Weil quadratic form in the zeta-family context. Section 3 introduces branch-locality formally and embeds it in the trinity. Sections 4, 5, and 6 recall the three canonical instances: Riemann (NE-A and NE-B), Selberg (Casimir operator), and Prime-Hub (Open Problem 5). Section 8 states and proves the boundary theorem. Section 9 introduces UCU formally and exhibits its role as the universal uniqueness statement across four programmes. Section 10 discusses Weil-form methods as the branch-transversal complement. Section 11 formulates the SGE conjecture and tests it on the Dedekind and Ihara families. Section 12 closes with the blockchain reading, which transports the essence of the trinity into an informatic register. Section 13 collects the rigor hierarchy, the consequences for Dirichlet, and the outlook on adjacent projects.

2 Setup

2.1 Zeta-type families

Definition 2.1 (Zeta-type family)**.** A *zeta-type family* is a tuple $(Z_X, \mathcal{T}_X, \mathcal{E}_X)$ consisting of

1. a meromorphic function $Z_X(s)$ on $\mathbb{C}$ with at most finitely many poles, non-trivial zeros $\{\rho_k = 1/2 + i\gamma_k^X\}$ on or near a critical line $\{\mathrm{Re}(s) = 1/2\}$, satisfying an Euler product on a right half-plane and a functional equation $Z_X(s) \leftrightarrow Z_X(1-s)$ up to a gamma factor;
2. a natural family $\mathcal{T}_X$ of bounded operators on a Hilbert space associated with the geometry/arithmetic of X ("transfer operators");
3. an explicit formula $\mathcal{E}_X$ relating test-function sums over zeros to prime/geodesic sums of the family.

Example 2.2 (Canonical instances)**.** (i) $X = \zeta$: $\mathcal{T}_\zeta = \{\text{prime-shift operators } D(r)\}$ with explicit formula of Weil type.
(ii) $X = Z_\Gamma$ for Γ co-compact Fuchsian: $\mathcal{T}_\Gamma = \{\text{geodesic transfer operators } T_{\ell_\gamma}\}$ with Selberg explicit formula.
(iii) $X = G_{\text{prime}}$: conjectural transfer operator on a prime-indexed graph; Open Problem 5 of [7].

2.2 The Hilbert-Pólya structure

Definition 2.3 (Branch-local Hilbert-Pólya)**.** A zeta-type family $(Z_X, \mathcal{T}_X, \mathcal{E}_X)$ is said to *possess Hilbert-Pólya structure*—written HP-BL(X) = YES—if there exists a self-adjoint operator $H_X : \mathcal{D}(H_X) \to \mathcal{H}_X$ on a Hilbert space $\mathcal{H}_X$ such that:

(HP1) $\mathrm{Spec}(H_X) = \{\gamma_k^X : k \in \mathbb{N}\}$, with multiplicities.

(HP2) H_X commutes with every $T \in \mathcal{T}_X$ in the sense of commuting spectral projections.

(HP3) H_X is *natural*: uniquely determined by the geometry or arithmetic of X, up to unitary equivalence.

If no such H_X exists, we write HP-BL(X) = NO; if its existence is open, HP-BL(X) = OPEN.

Remark 2.4 (Informal intuition)*.* Hilbert and Pólya's original suggestion was that the Riemann hypothesis would follow from exhibiting a natural self-adjoint operator "in Nature" whose eigenvalues coincide with the imaginary parts of the zeta zeros. The three conditions (HP1)–(HP3) formalise this informal requirement: spectral identity, commutation with the natural transfer operators of the family, and a naturality condition that prevents tautological constructions (e.g. the trivial multiplication operator on ℓ^2 indexed by the zeros themselves).

2.3 The Weil quadratic form as branch-transversal architecture

For any zeta-type family $(Z_X, \mathcal{T}_X, \mathcal{E}_X)$, the Weil explicit formula [41] induces a quadratic form Q_X on $L^2(\mathbb{R})$ of the schematic shape

$$Q_X[f] = \sum_k 2\pi |\widehat{f}(\gamma_k^X)|^2 = W_\infty^X[f] + (\text{prime/geodesic side}). \tag{1}$$

Positivity of Q_X on a suitable test-function class is equivalent to the Riemann hypothesis for Z_X, independently of the existence of any Hilbert-Pólya operator. This is the branch-transversal content of the Weil architecture: it applies in the HP-BL-YES, -NO, and -OPEN cases uniformly.

2.4 Three-component decomposition of RH-type statements

A further structural refinement, following [2], isolates three conceptually distinct layers inside every RH-type statement for a zeta-type family X. Writing $\mathrm{RH}(X)$ for the positivity of Q_X on the admissible test-function class, one has the *logical* decomposition (read as a conjunction of necessary-and-sufficient conditions, not as an arithmetic sum)

$$\mathrm{RH}(X) \iff \underbrace{\mathrm{GBC}(X)}_{\text{Galerkin-Bridge}} \wedge \underbrace{C^{(2)}(X)}_{\text{eigenvector realisation}} \wedge \underbrace{\mathrm{Hurwitz}(X)}_{\text{universal limit}}. \tag{2}$$

(2) is a *schema*, not a theorem: it records which three layers every RH-type programme must supply, and labels them as the targets of distinct technical sub-programmes. The precise form-domain conventions under which each component is defined are fixed in the individual instances of Sections 4–6.

The three components are as follows.

(GBC) *Galerkin-Bridge component.* The algebraic-spectral step: for every cutoff $\lambda > 1$ one constructs a finite-dimensional Galerkin approximation $Q_X^{(N,\lambda)}$ of Q_X on a logarithmic host space (cosine/sine basis on $[-\log\lambda, \log\lambda]$ in the arithmetic case), shows positivity/negativity of the approximation via interval arithmetic or combinatorial arguments, and transfers the result to the continuous operator via Mosco convergence [2, 25]. This layer is where the v2.1 Direct Frontier Dominance proof [6] lives for $X = \zeta$.

$\boldsymbol{C^{(2)}}$ *Eigenvector realisation component.* The analytic step: one shows that the Galerkin eigenvectors $f_\star^{(N)}$ converge, in a sufficiently strong topology, to a genuine eigenfunction $f_\star$ of the continuous operator. For the Riemann case this is precisely the Paley–Wiener control required in Connes' programme [24, 25] and flagged in [6] as the deepest remaining analytic ingredient before the Zookeeper closure. In the current CORE status, [14] closes this $C^{(2)}$/MS2 step in the CCM/Fourier model and supplies the microcluster normal form used below. For Selberg this step is automatic (compact resolvent of the Laplacian on $\Gamma\backslash\mathbb{H}$).

(Hurwitz) *Universal limit component.* The final functorial step: Hurwitz' theorem on locally uniform limits of non-vanishing holomorphic functions promotes the family of finite-level statements to the limit on the critical line. This component is independent of X: it is the same analytic lemma that enters every λ-cutoff RH argument, which is why we list it separately as "universal".

Remark 2.5 (Where each FST branch specialises)*.* The three-component decomposition makes visible *where* each branch of FST concentrates its technical effort:

- The **v2.1 Even Dominance programme** [6] carries the GBC layer unconditionally for $X = \zeta$ and reduces $\mathrm{RH}(\zeta)$ to $C^{(2)}(\zeta)$.
- The **Connes programme** [24, 25] is the natural home of the $C^{(2)}$ step, formulated semilocally as an analytic condition on prolate-spheroidal eigenfunction asymptotics; in the Zookeeper paper this step is closed in the CCM/Fourier branch by the three-lemma microcluster argument.
- The **Obstruction-Classifier programme** [26] diagnoses on which (GBC, $C^{(2)}$) pair a given problem is hard: easy GBC + Zookeeper-closed $C^{(2)}$ (Riemann), twisted-transfer $C^{(2)}$ (Dirichlet), easy both (Selberg, automorphic L on GL_n), open GBC (Ihara generic, Yang–Lee, p-adic L).
- The **Hurwitz** layer is invariant across all these branches; when we "take the limit" to the continuous critical line, the same lemma does the work.

In particular, the trinity of Section 9—UCU + SGE + Weil-form transversality—acts on the GBC layer (UCU supplies uniqueness of the Galerkin ground state, SGE classifies the admissible transfer algebras, Weil-form transversality is the architecture carrying the Galerkin approximation). $C^{(2)}$ and Hurwitz are transversal to this trinity; they enter *after* the trinity has fixed the ground state.

Remark 2.6 (The boundary-theorem view). Theorem 8.1 (HP-BL) classifies zeta-type families by their HP status. The three-component decomposition refines this by telling us *which component determines the classification*:

- HP-BL = YES (Selberg, Ihara Petersen, automorphic): $C^{(2)}$ is automatic (compact resolvent); the HP operator is the natural Laplacian.
- HP-BL = NO (Riemann, Dedekind $\mathbb{Q}(\sqrt{-5})$, Dirichlet): HP-BL failure/no-construction is not identical with an open $C^{(2)}$ status. For Riemann the Zookeeper paper closes $C^{(2)}$/MS2 in the CCM/Fourier branch; for Dirichlet and Dedekind the corresponding twisted transfer remains to be tested. The Weil-form route remains branch-transversal.
- HP-BL = OPEN (Prime-Hub): the explicit OP5 graph construction and its boundary-operator interpretation remain open; this is no longer read as an open Riemann $C^{(2)}$ blocker after Zookeeper.

3 Branch-locality as a concept

Branch-locality refers to the observation that a structural property of zeta-type families can hold for one family and fail for a closely related one. Before we state the boundary theorem, we make the concept precise.

3.1 Three modes of HP-BL

Definition 3.1 (HP-BL status trichotomy)**.** For a zeta-type family X in the sense of Definition 2.1, we say that

- HP-BL(X) = YES if there exists an operator H_X satisfying (HP1)–(HP3) of Definition 2.3;
- HP-BL(X) = NO if one can prove, or has established under a specific conjecture, that no such H_X exists;
- HP-BL(X) = OPEN if no candidate has been exhibited *and* no exclusion has been proved, possibly subject to identified obstructions.

The status NO may be *unconditional* or *conditional*. In the present work, the Riemann case will be NO conditional on Conjecture 4.4 (the extension of NE-B from finite-dimensional Galerkin truncations to the full space). The conjecture is conjectural but structurally precise: an explicit open statement whose resolution would either confirm or refute the NO-classification.

3.2 Naturality (HP3): the operational test

The naturality clause (HP3) of Definition 2.3 is the subtle one. Without it, a trivial "HP operator" always exists: on the Hilbert space $\ell^2(\{\gamma_k\})$ one can simply define H_X as multiplication by γ_k. Such an ad hoc H_X satisfies (HP1) tautologically but fails to explain anything. We therefore require that H_X be *natural* in the following operational sense.

Definition 3.2 (Natural HP operator)**.** A self-adjoint operator H_X on $\mathcal{H}_X$ is called *natural* for the zeta-type family X if:

(N1) there is a natural symmetry group G_X acting on X (via its geometry, arithmetic, or automorphism structure) under which H_X is invariant;

(N2) the Hilbert space $\mathcal{H}_X$ and the embedding $\mathcal{T}_X \hookrightarrow \mathcal{B}(\mathcal{H}_X)$ are G_X-equivariant;

(N3) H_X is uniquely determined by (HP1)–(HP2) and (N1)–(N2) up to unitary equivalence commuting with the G_X-action.

In Selberg's case, the symmetry group is $\Gamma\backslash\mathrm{PSL}_2(\mathbb{R})$, the Hilbert space is $L^2(\Gamma\backslash\mathbb{H})$, and the Laplace-Beltrami operator is uniquely characterised as the second-order Γ-invariant differential operator up to a scalar (§5). In the Riemann case, the prime-shift family $\{D(r) : r \in (0, 2)\}$ does not descend from any natural finite-dimensional Lie group action of the form required by (N1); Theorem 4.3 establishes that no non-scalar symmetric operator even commutes with all $D_N(r)$ at the tested truncations. The Prime-Hub graph, if constructed, would need to supply an explicit G_{prime}, which is part of Open Problem 5.

3.3 Why not a uniform theorem across all zeta-type families?

A plausible naive hope is that HP-BL would be *uniformly* YES across zeta-type families: every L-function would carry its own natural H_X. The boundary theorem below is a precise quantification of the failure of this hope: even the three canonical instances $\{\zeta, Z_\Gamma, G_{\mathrm{prime}}\}$ already realise all three values $\{\mathrm{NO}, \mathrm{YES}, \mathrm{OPEN}\}$. Any uniform claim across zeta-type families is therefore false.

A subtler hope is that HP-BL = YES would be *generic*, with the Riemann case somehow exceptional. Theorem 4.3 disallows even this reading on the tested Galerkin truncations: the absence of a common symmetry among prime-shift operators is an explicit structural feature of the multiplicative prime structure, not a pathology. Conversely, the presence of a Lie-group-invariant Laplacian in Selberg's setting is a consequence of the specific geometry of $\Gamma\backslash\mathbb{H}$, not of zeta-analyticity per se. The branch-locality thesis is that HP-BL-status is determined by the *algebraic substrate* of the family, independently of its zeta-analytic features.

3.4 A three-class taxonomy of the zeta zoo

The branch-locality trichotomy HP-BL $\in \{\mathrm{YES}, \mathrm{NO}, \mathrm{OPEN}\}$ of Definition 3.1 classifies families by the *existence* of a natural Hilbert-Pólya operator. A second, largely orthogonal classification captures the *type of spectrum* that such an operator, if it exists, admits. Three classes emerge:

Class	**Transfer algebra**	**Typical HP**	**Spectrum**	**Zoo examples**
Arithmetic zetas	Prime/prime-ideal semigroup	NO	Discrete (zeros)	Riemann, Dirichlet, Dedekind
Geometric zetas (compact)	Discrete Γ on compact quotient	YES	Discrete (Laplace)	Selberg, Ihara
Flow zetas (non-compact)	Continuous Lie group on non-compact space	YES*	Continuous + resonances	CRM, Ruelle-Anosov, BH-QNM

* realised in the resonance sense: the HP operator (Casimir of the acting Lie group) has absolutely continuous spectrum, and the zeta zeros correspond to scattering resonances in the lower half plane (cf. [16]).

The three classes are orthogonal to the HP-BL-trichotomy: both Selberg and CRM sit on the HP-BL = YES-side of the SGE dichotomy but belong to different spectral classes. Riemann and Dedekind share the arithmetic class but remain separate from the geometric and flow classes entirely. We will exhibit one canonical member of each class in Sections 4–7, and give the complete boundary theorem in Section 8.

3.5 Relation to Connes' adelic framework

In Connes' programme [23, 24], the relevant "space" on which a putative H_ζ acts is an adelic quotient $\mathbb{A}_K/K^\times$, and the natural symmetry is the GL_1-action of idele classes. Our formulation Definition 3.2 does not exclude this architecture: if Connes' programme succeeds, his operator would satisfy (N1)–(N3) with $G_X = \mathrm{GL}_1(\mathbb{A}_K/K^\times)$, and would constitute an explicit HP-BL = YES certificate for $\zeta(s)$ in the adelic Hilbert space. That possibility is currently open: it would coexist with Theorem 4.3 at finite dimensions because the adelic natural G_X is not visible in the Galerkin truncation. Our boundary theorem therefore concerns Hilbert-Pólya structure with respect to the *prime-shift* transfer operators $\mathcal{T}_\zeta$, which is the relevant structure for the Weil quadratic form and the Even Dominance proof of [6].

The sections that follow exhibit one canonical instance of each HP-BL mode.

4 Instance A: Riemann (NE-A and NE-B)

4.1 Non-existence of absolute total positivity (NE-A)

We quote the following theorem from [6]:

Theorem 4.1 (NE-A, [6, Thm. NE-A])**.** *The Fourier multiplier $M(\xi) = -2\,\mathrm{Re}[\zeta'/\zeta(\frac{1}{2}+i\xi)]$ of the prime-shift operator takes negative values on a set of positive Lebesgue measure in every interval $[-L, L]$ with $L > \gamma_1$ (the first positive zeta ordinate). Consequently, the prime-shift operator is not absolutely totally positive (*PF_∞*).*

Remark 4.2. Empirically (cf. [6, §4.2]), sampling $M(\xi)$ on a uniform grid of 10^4 points in $\xi \in [0, 100]$ with ε-neighbourhoods of the first 29 ordinates excluded yields approximately 49% of points with $M(\xi) < 0$. This illustrates the theorem; it does not replace the structural argument.

4.2 No universal commuting operator (NE-B)

Theorem 4.3 (NE-B, [6, Thm. NE-B])**.** *For every Galerkin truncation $N \leq 15$ and every sufficiently dense sample of normalised shifts $r_1, \ldots, r_k \in (0, 2)$, the only symmetric $N \times N$ matrix commuting with all prime-shift matrices $D_N(r_j)$ is a scalar multiple of the identity.*

Conjecture 4.4 (NE-B-full, [6, Conj. 4.3-full])**.** *Theorem 4.3 extends to $N = \infty$: no non-trivial universal commuting operator exists for the Riemann prime-shift family.*

4.3 Consequence for HP-BL(ζ)

Theorems 4.1 and 4.3 jointly exclude the two most natural routes to (HP1)–(HP3) for the Riemann zeta family. No absolutely totally positive prime shift operator can serve as H_ζ (Route A), and no universal commuting self-adjoint symmetric operator exists at any tested Galerkin truncation (Route B). Under Conjecture 4.4, HP-BL(ζ) = NO.

Remark 4.5 (Paper II's RH proof does not require HP structure). The Even Dominance proof of the Riemann hypothesis given in [6] proceeds by prime-weighted summation (Shift Parity Lemma) plus frontier-dominance arguments using the Weil quadratic form. Neither step requires (HP1)–(HP3). This is consistent with the branch-locality thesis: even though HP-BL(ζ) = NO (conditional on Conjecture 4.4), the Weil-form architecture still yields the desired spectral conclusion.

Remark 4.6 (Documentation status of the Direct Frontier proof). We cite [6] as providing the Direct Frontier Dominance proof unconditionally. A systematic critique of the trilogy [6, 7]—covering (K1) the global extrapolation from the $\lambda = 100$ CAP baseline, (K2) the transparency of the PNT/Mertens constants $C_{\text{shift}}, C_{\text{norm}}$, (K3) the Galerkin convergence to the infinite-dimensional Weil operator, (K4) the implication-versus-equivalence structure of the reduction chain A1–A8, (K5) the precise characterisation of the form domain of Q_ζ, and (K6) overall reviewer ergonomics—was carried out in a companion review [28]. The verdict was that all six critiques are *partially valid but non-fatal*: the proof architecture is substantively sound, but a revised trilogy (RH v2.2) would sharpen (i) the per-row equivalence/implication markers of the reduction chain, (ii) an explicit Galerkin convergence lemma extending the $D_3(r)$-matrix argument to the full Weil operator, and (iii) explicit interval-arithmetic certificates for $C_{\text{shift}}, C_{\text{norm}}$. These refinements do not affect the boundary theorem or the SGE classification used in the present paper; they strengthen the RH v2.1 documentation without changing its technical content.

5 Instance B: Selberg (Casimir operator)

5.1 The Laplace-Beltrami as explicit Hilbert-Pólya operator

Let $\Gamma \subset \mathrm{PSL}_2(\mathbb{R})$ be a co-compact Fuchsian group and $\mathcal{F} = \Gamma\backslash\mathbb{H}$ the associated compact hyperbolic surface. The Selberg zeta function $Z_\Gamma(s)$ has non-trivial zeros $\rho_n = \frac{1}{2} + ir_n$ related to the spectrum of the Laplace-Beltrami operator $\Delta_{\mathcal{F}}$ on $L^2(\mathcal{F})$ via

$$\Delta_{\mathcal{F}}\,\phi_n = \lambda_n \phi_n, \qquad \lambda_n = \tfrac{1}{4} + r_n^2, \tag{3}$$

with finitely many exceptions from small eigenvalues [40].

Theorem 5.1 (Selberg)**.** *The operator $H_{Z_\Gamma} := \sqrt{\Delta_{\mathcal{F}} - 1/4}$ satisfies (HP1)–(HP3) of Definition 2.3 for the zeta-type family $(Z_\Gamma, \mathcal{T}_\Gamma, \mathcal{E}_\Gamma)$. In particular,* HP-BL$(Z_\Gamma)$ = YES*.*

Sketch. (HP1) follows from (3). (HP2) follows from Γ-invariance: the Laplace-Beltrami operator commutes with every isometry-pullback and hence with every geodesic transfer operator. (HP3) follows from the uniqueness of the Laplacian as the second-order Γ-invariant differential operator. The spectral square root in the exceptional-eigenvalue regime is handled by the standard functional calculus on the absolutely continuous part of $\mathrm{Spec}(\Delta_{\mathcal{F}})$. □

5.2 Selberg trace formula: the explicit mechanism

The Selberg trace formula [40] schematically relates spectral and geometric data:

$$\sum_n \widehat{h}(r_n) = \frac{\mathrm{vol}(\mathcal{F})}{4\pi} \int h(r)\, r \tanh(\pi r)\, dr + \sum_{\{\gamma\}} \frac{\ell(\gamma_0)}{2\sinh(\ell(\gamma)/2)}\, g(\ell(\gamma)), \tag{4}$$

where r_n are the spectral parameters of $\Delta_{\mathcal{F}}$, the second sum runs over primitive closed geodesics $\{\gamma_0\}$ and their iterates, and h, g are a standard test-function pair related by Fourier transform. The geodesic side of (4) is the *multiplicative-geometric analogue* of the prime side of the Riemann–Weil explicit formula.

The Γ-invariance of $\Delta_{\mathcal{F}}$ is the *structural reason* that the geodesic side has this closed form: geodesics γ are orbits under the $\mathrm{PSL}_2(\mathbb{R})$ action modulo Γ, and the transfer operators T_{ℓ_γ} are conjugate to Γ-translations along γ. The Casimir of the ambient $\mathrm{PSL}_2(\mathbb{R})$ acts as the second-order generator of these geodesic flows and therefore commutes with every T_{ℓ_γ} simultaneously. This is the concrete verification of (N1)–(N3) of Definition 3.2.

5.3 What distinguishes Selberg from Riemann

The Riemann zeta family admits a formal analogue of (4) in the shape of the Weil explicit formula:

$$\sum_k h(\gamma_k^\zeta) = W_\infty[h] - 2\sum_{p,m} \frac{\log p}{p^{m/2}}\,\widehat{h}(m\log p), \tag{5}$$

with archimedean weight W_∞ encoding the gamma factor. The prime side of (5) is *multiplicative*: it is indexed by the semigroup of prime powers, not by orbits of a Lie group action. There is no natural continuous group whose discrete orbits reproduce $\{p^m\}$, and Theorem 4.3 confirms this structurally: no non-scalar symmetric operator commutes with the prime-shift operators at the tested Galerkin truncations.

The precise structural difference between (4) and (5) is therefore:

Feature	Selberg (HP-BL YES)	Riemann (HP-BL NO*)
Indexing set	closed geodesics $\{\gamma\}$	prime powers $\{p^m\}$
Group action	$\mathrm{PSL}_2(\mathbb{R})$ on $\mathbb{H}$	none (NE-B)
Commuting Hamiltonian	$\Delta_{\mathcal{F}}$ = Casimir	none at $N \leq 15$
Length structure	continuous $\ell(\gamma) \in \mathbb{R}_{>0}$	discrete $\log p \in \mathbb{R}_{>0}$

The multiplicative arithmetic of the primes is *not the orbit structure of any natural continuous symmetry group.* This is the content of branch-locality: the algebraic substrate of the transfer-operator family decides the HP-BL status, independently of the zeta-analytic similarity between (4) and (5).

6 Instance C: Prime-Hub (Open Problem 5)

Open Problem 5 of [7], hereafter OP5, calls for an explicit topologically mixing directed graph $G_{\text{prime}} = (V, E)$ (or equivalently a flow or dynamical system) satisfying five conditions:

(OP5.1) Primitive periodic orbits $\{\gamma_p\}$ are in bijection with the primes.

(OP5.2) $\ell(\gamma_p) = \log p$ for every prime p.

(OP5.3) The associated transfer operator $\mathcal{L}_s$ has Fredholm determinant $\det(I - \mathcal{L}_s)^{\text{vac}} = 1/\zeta(s)$.

(OP5.4) The spectral radius of $\mathcal{L}_s$ is less than 1 for $\mathrm{Re}(s) > 1/2$.

(OP5.5) Eigenvalue 1 occurs exactly at the non-trivial zeros of $\zeta(s)$.

No construction simultaneously satisfying (OP5.1)–(OP5.5) is known since the informal Hilbert-Pólya suggestion of 1914 [19]. The programme is reviewed and surveyed in [24]; Connes' *Letter Through Time* (*ibid.*, final sections) proposes a trace-formula-based convergence strategy from finite to infinite Euler products, but a constructive G_{prime} is not exhibited.

6.1 Obstruction C1: Weyl-law mismatch

Proposition 6.1 (Weyl-law mismatch)**.** *Let H be the Laplace-Beltrami operator of a smooth compact d-dimensional Riemannian manifold. Then its eigenvalue counting function satisfies Weyl's law*

$$N_H(T) := \#\{\lambda \in \mathrm{Spec}(H) : \lambda \le T^2\} \;\sim\; \frac{\omega_d \,\mathrm{vol}(M)}{(2\pi)^d}\, T^d \qquad (T \to \infty), \tag{6}$$

with ω_d the volume of the unit ball in $\mathbb{R}^d$. The Riemann-zero counting function is

$$N_\zeta(T) := \#\{\gamma_k^\zeta : 0 < \gamma_k^\zeta \le T\} \;\sim\; \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T) \tag{7}$$

(classical Riemann-von Mangoldt, cf. [43, §9.4]). The asymptotic rates (6) and (7) are incompatible for every integer $d \ge 1$. In particular, no classical geometric Laplacian on a smooth compact manifold can realise $\mathrm{Spec}(H_\zeta)$.

Proof. The power-law (6) contrasts with the logarithmic density (7): for any $d \ge 1$, the ratio $N_\zeta(T)/N_H(T) \to 0$ (if $d \ge 2$) or $\to \infty$ as $T \to \infty$ (if $d = 1$), so no choice of d and $\mathrm{vol}(M)$ makes (6) match (7). Consequently, any candidate H_ζ with $\mathrm{Spec}(H_\zeta) = \{\gamma_k^\zeta\}$ cannot be the Laplacian of a smooth compact Riemannian manifold. □

Remark 6.2. Proposition 6.1 rules out smooth compact manifold Laplacians but not all possibilities: non-compact spectra, hyperbolic-like volume growth (Berry's chaotic quantisation [21, 22]), or adelic spaces [23, 24] remain candidate architectures. The obstruction is therefore a restriction on the *type* of geometric substrate.

6.2 Obstruction C2: Trace-class regularity

Proposition 6.3 (Trace-class regularity)**.** *The Fredholm determinant* $\det(I - \mathcal{L}_s)^{\mathrm{vac}}$ *in requirement (OP5.3) is well-defined only if $\mathcal{L}_s$ is trace-class on its ambient Hilbert space. For a graph transfer operator whose primitive orbits have logarithmic lengths $\ell(\gamma_p) = \log p$, the summability condition*

$$\sum_{p,m} \frac{1}{p^{m/2}} \cdot |\lambda_p^m| \;<\; \infty \tag{8}$$

(with λ_p the eigenvalue of $\mathcal{L}_s$ along the primitive cycle γ_p) must hold uniformly on the regime where (OP5.4) demands spectral radius less than 1*. No explicit construction simultaneously satisfying (OP5.1)–(OP5.5) and (8) has been exhibited; this is known in the Geiger tree as obstruction K4 of [1].*

Remark 6.4. The difficulty behind Proposition 6.3 is that (OP5.4) and (8) pull in opposite directions: (OP5.4) constrains the eigenvalues λ_p from above by 1, while (8) constrains their *sum* to be absolutely convergent. For a "generic" graph with infinitely many primitive orbits, natural operator models give either violation of (OP5.4) (eigenvalues pile up near 1) or violation of (8) (slow decay). Threading this needle is the substantive content of OP5.

6.3 Obstruction C3: NE-A compatibility

Proposition 6.5 (NE-A compatibility)**.** *Any Hilbert-Pólya candidate H_{prime} for ζ implicitly determines the Fourier multiplier $M(\xi) = -2\,\mathrm{Re}[\zeta'/\zeta(1/2 + i\xi)]$ of the prime-shift operator via the trace formula of $(H_{\mathrm{prime}}, \mathcal{L}_s)$. By Theorem 4.1, $M(\xi) < 0$ on a set of positive Lebesgue measure. Consequently, H_{prime} cannot arise as a non-negative self-adjoint operator with non-negative Fourier symbol on smooth geometries; it must carry intrinsic sign structure in its spectral transform.*

Sketch. A Hilbert-Pólya candidate must reproduce the Weil explicit formula (Section 2.3) via its trace. The kernel of the trace formula is precisely $M(\xi)$. Standard Laplacians on smooth geometry produce non-negative Fourier multipliers; the sign of $M(\xi)$ on a positive-measure set therefore rules out such Laplacians. Construction must use structures with inherent sign change (e.g. pseudo-differential, or noncommutative adelic). □

Remark 6.6. Obstructions C1 (smooth geometry ruled out) and C3 (non-negative symbols ruled out) are independent but mutually reinforcing: C1 rules out manifold architecture by density, C3 rules out manifold architecture by symbol sign. Even if one workaround (e.g. hyperbolic chaotic quantisation) circumvents C1, it must still engineer a symbol satisfying C3.

6.4 Consequence

The combination of Propositions 6.1, 6.3, and 6.5 identifies three structural obstructions that any explicit Prime-Hub construction must navigate. None of the three is individually a proof of impossibility; together, they delimit the narrow parameter space in which a valid G_{prime} could exist. Consequently, HP-BL(G_{prime}) = OPEN, with obstructions C1–C3 as explicit guides for any future construction. Connes' *Letter Through Time* [24] exemplifies the kind of non-smooth, noncommutative architecture that could in principle navigate all three; whether the programme closes remains open.

7 Instance D: CRM as Flow-Zeta (FST-Cosmology bridge)

Instances A, B, C exhibit the arithmetic and geometric-compact classes of the three-class taxonomy of Section 3.4. The flow-zeta class is populated by a concrete object from the physical side of FST: the Cooperative Renormalization Model (CRM) of cosmological dark energy [3]. Its mathematical structure is developed in full detail in [13]; we summarise the key points here.

7.1 CRM as a one-parameter Lie group

The CRM composition law governing dark-energy evolution is

$$x \oplus y = \frac{x+y}{1+xy}, \qquad x, y \in (-1, 1). \tag{9}$$

Setting $x = \tanh u$, $y = \tanh v$ gives $x \oplus y = \tanh(u+v)$, so $(x, \oplus)$ is isomorphic to the additive group $(\mathbb{R}, +)$. The composition is therefore the Lorentz-boost addition, and the relevant transfer algebra is a *one-parameter Lie group* realised as a subgroup of $\mathrm{PSL}_2(\mathbb{R})$ acting on the hyperbolic plane $\mathbb{H}$.

Proposition 7.1 (CRM transfer algebra)**.** *For the CRM family $X = \mathrm{CRM}$, the transfer algebra is $(I_{\mathrm{CRM}}, \cdot) = (\mathbb{R}, +)$, a locally compact abelian Lie group. Consequently, under Conjecture 11.1,* HP-BL(CRM) = YES*.*

7.2 The natural Hilbert-Pólya operator and its spectrum

The Casimir of $\mathfrak{sl}_2(\mathbb{R})$ restricted to the boost-generating component is, up to equivalence, the Laplace-Beltrami operator $\Delta_{\mathbb{H}}$ on the hyperbolic plane [16]. $\Delta_{\mathbb{H}}$ satisfies (HP1)–(HP3) of Definition 2.3 in the form appropriate for non-compact underlying geometry: its absolutely continuous spectrum is $[1/4, \infty)$, and it commutes with the full $\mathrm{PSL}_2(\mathbb{R})$-action, hence with the CRM flow in particular.

The *zeros* of ζ_{CRM} are, in this setting, *scattering resonances*: poles of the meromorphic continuation of the resolvent $(\Delta_{\mathbb{H}} - s(s-1))^{-1}$ through the essential spectrum into the lower half plane. On asymptotically hyperbolic backgrounds (e.g. de Sitter, Schwarzschild–de Sitter) these resonances are precisely the *quasinormal modes* of the relevant wave equation [16]. This realises HP-BL = YES in the resonance sense of Section 3.4.

7.3 The Ruelle flow-zeta of CRM

Following [16], the appropriate analytic object is the Ruelle-Dyatlov-Zworski zeta

$$\zeta_{\mathrm{CRM}}(s) = \prod_{\gamma \in \mathcal{P}_{\mathrm{CRM}}} \frac{1}{1 - e^{-s\,\ell(\gamma)}}, \tag{10}$$

where $\mathcal{P}_{\mathrm{CRM}}$ indexes the primitive periodic orbits of the CRM flow on an asymptotically hyperbolic background, and $\ell(\gamma)$ is the orbit length in the Lorentz-boost metric. The zeros of $\zeta_{\mathrm{CRM}}(s)$ are the resonances described above.

7.4 CRM as a bridge species

CRM occupies a distinctive position in the zeta zoo: it is simultaneously a flow-zeta entry in the mathematical zoo (this section) and a Pattern A instantiation in the physical RFEP ecosystem [3] (see also the narrative closure in Section 1.4). Together with Riemann (historical origin), BSD, Hodge, and P vs NP, CRM forms the fifth *bridge species* of the programme—and the first to open the flow-zeta column on both sides of the FST twins.

7.5 The CRM saturation theorem as a classification statement

A structural remark is in order. The CRM V saturation theorem [15] asserts that any dynamical system on $(-1, 1)$ satisfying four axioms—existence of a scalar relaxation variable, existence of a dissipative term, existence of a stable saturation value, and monotone smooth dynamics—necessarily realises the tanh saturation profile. Read as a characterisation of the flow-zeta class, this is not merely a statement about one cosmological model; it is a *classification statement* for the class of IR-asymptotics compatible with functional stability on the boost interval.

This is structurally analogous, on the flow side, to the Hurwitz/ compactness characterisation used in the arithmetic branch: just as the Weil quadratic form detects stability across branches, CRM V delineates the admissible asymptotic class on the flow-noncompact branch. In particular, several candidate UV completions may realise distinct selections from this saturation class; the classification does not depend on any individual realisation.

This observation matters for two reasons. First, it frames CRM not as a bespoke cosmological proposal but as the IR sector of a branch-local structural theorem. Second, it predicts that any UV theory whose IR limit passes the four CRM V axioms (e.g. a GUT-level geometric model, an RG-flow construction, or an open-system holographic construction) lands in the same saturation class—modulo quantitative parameter matching. Making this UV/IR compatibility test operational is an open programme within FST-Cosmology.

8 The Boundary Theorem

Theorem 8.1 (Branch-Locality of Hilbert-Pólya)**.** *The* HP-BL*-property is not constant across zeta-type families. When combined with the three-class taxonomy of Section 3.4, the zoo exhibits four canonical instances plus two empirical confirmations:*

Family	HP-BL	*Class*	*Evidence*
Riemann $\zeta(s)$	*NO**	*arithmetic*	*Thms. 4.1, 4.3; Conj. 4.4*
Selberg $Z_\Gamma(s)$	*YES*	*geometric-compact*	*Thm. 5.1*
Prime-Hub G_{prime}	*OPEN*	—	*Props. 6.1–6.5*
CRM (Dark Energy)	*YES*	*flow-noncompact*	*Prop. 7.1, [16]*
Dedekind $\mathbb{Q}(\sqrt{-5})$	*NO*	*arithmetic*	*§11.2, [11]*
Ihara $\zeta_{Petersen}$	*YES*	*geometric-compact*	*§11.3, [12]*

* *conditional on Conjecture 4.4.*

Proof. The three rows are the contents of Sections 4–6. The values YES, NO, and OPEN are distinct, so the HP-BL-property is non-constant. The theorem is the formal summary of this non-constancy. ☐

Corollary 8.2 (Implication for the Riemann hypothesis)**.** *Any unconditional proof of the Riemann hypothesis that relies on (HP1)–(HP3) for ζ must first refute Conjecture 4.4. Alternatively, Weil-form methods (Section 10) yield an unconditional route that does not require the Hilbert-Pólya structure; this is realised in [6].*

9 Universal Convexity Uniqueness (UCU)

This section isolates the uniqueness principle that operates, in precisely the same abstract form, in all programmes of the functional-stability landscape considered in this paper. We call it *Universal Convexity Uniqueness* (UCU). It is the mathematical lemma underlying what is called *Pattern A* in the physical formulation (RFEP; [3]) and its dynamical partner *Dissipative Selection* (DS1–DS3). UCU is a single technical statement; its universality lies in how many seemingly unrelated objects realise it.

9.1 The UCU lemma

Lemma 9.1 (UCU)**.** *Let $(\mathcal{H}, \langle\cdot,\cdot\rangle)$ be a real Hilbert space, let $\mathcal{A} \subset \mathcal{H}$ be a closed convex admissible class, and let $J : \mathcal{A} \to \mathbb{R}$ satisfy*

(C) Strict convexity: *for all $u, v \in \mathcal{A}$ with $u \neq v$ and all $t \in (0,1)$,*

$$J(tu + (1-t)v) < t\,J(u) + (1-t)\,J(v);$$

(L) Strong lower semicontinuity: *if $u_n \to u$ strongly in $\mathcal{H}$ with $u_n, u \in \mathcal{A}$, then $J(u) \leq \liminf_{n\to\infty} J(u_n)$;*

(K) Coercivity on $\mathcal{A}$: *$J(u) \to +\infty$ whenever $\|u\| \to \infty$ within $\mathcal{A}$.*

Then J has a unique minimiser $u_\star \in \mathcal{A}$.

Proof. Let $\{u_n\}$ be a minimising sequence. By (K) it is bounded, so it admits a weakly convergent subsequence $u_{n_k} \rightharpoonup u_\star$ with $u_\star \in \mathcal{A}$ by weak closedness of closed convex sets. Convexity (C) plus strong lower semicontinuity (L) together imply weak lower semicontinuity of J (Mazur's lemma; see [29, Prop. II.1.2]), so $J(u_\star) \leq \liminf J(u_{n_k}) = \inf_{\mathcal{A}} J$, and $u_\star$ is a minimiser. Strict convexity (C) excludes a second minimiser: if $v_\star \neq u_\star$ also realised the minimum, the midpoint $\frac{1}{2}(u_\star + v_\star) \in \mathcal{A}$ would strictly decrease J, contradicting minimality. ☐

The content of Lemma 9.1 is classical, but the scope of its role in the present programme is not: it is the *same* lemma, with different $(\mathcal{H}, \mathcal{A}, J)$, that yields the unique saturation state in each of the five realisations listed below.

9.2 The parallel realisations

Table 1 exhibits the five instances. Each column shows which object plays the role of the Hilbert space $\mathcal{H}$, the admissible class $\mathcal{A}$, and the functional J. The last row records the unique minimiser $u_\star$ and its physical or arithmetic interpretation.

Programme	**Functional J on class $\mathcal{A}$**	**Unique minimiser $u_\star$**
Riemann Even Dominance [6]	$J(f) = \langle f, Q_\zeta f\rangle/\|f\|^2$ on $\mathcal{A} = L^2_{\text{even}}([-L, L]) \setminus \{0\}$; convexity on even test functions reflects the positive-definiteness of the Weil quadratic form after Direct Frontier Dominance.	The ground state $f_\star$ realising the spectral gap; its existence and uniqueness are exactly the even–odd dominance statement.
Selberg zeta (Theorem 5.1)	$J(\phi) = \int_{\Gamma\backslash\mathbb{H}} \|\nabla\phi\|^2$ on $\mathcal{A} = \{\phi \in H^1_0(\mathcal{F}) : \|\phi\|_{L^2} = 1\}$; the Rayleigh quotient is strictly convex on the sphere after mod-out of the S^1-phase.	The first Laplace–Beltrami eigenfunction ϕ_1, whose Casimir eigenvalue supplies the spectral Hilbert–Pólya content.
Turbulence / Kolmogorov K41	$J(u) = \int \|\nabla u\|^2 - \epsilon \int u$ on $\mathcal{A}$ = admissible divergence-free velocity fields with fixed energy dissipation ϵ; strict convexity holds in the inertial range via the exponent-2/3 scaling.	The unique scaling profile producing the $k^{-5/3}$ energy cascade [5].
Yang–Mills Gibbs measure	$J(A) = S_{\text{YM}}(A) = \frac{1}{4g^2}\int F \wedge \star F$ on $\mathcal{A} = \mathcal{A}_{\text{flat}}/\mathcal{G}$, the moduli space of gauge-equivalent connections; strict convexity on the quotient follows from the Coulomb gauge condition.	The vacuum sector achieving the mass gap [4].
CRM (flow-zeta, §7)	$J(\text{rapidity}) = V_{\text{eff}}(\text{rapidity})$, an effective dark-energy potential on $(-1, 1)$ with boundary coercivity at $\|x\| \to 1$ built in by the Lorentz-boost composition $\oplus$; strict convexity is the structural input of the saturation axioms [15].	The unique saturated boost state, producing the observed tanh profile in $w(z)$.

Table 1: UCU instances. Same lemma, different $(\mathcal{H}, \mathcal{A}, J)$. In each row, strict convexity plus coercivity of J on $\mathcal{A}$ yields a unique minimiser; the minimiser is the saturation state of the programme.

9.3 UCU and Pattern A / RFEP: the translation

The physical face of UCU in the Renormalized Free Energy Principle framework [3] is *Pattern A*: functional positivity under a gauge constraint. The correspondence is exact:

- **Pattern A (PA1).** A target functional J (free energy, Weil form, Rayleigh quotient, K41 action, Yang–Mills action) is positive-definite on a gauge-restricted state space.
  $\longleftrightarrow$ *UCU hypothesis (C).*
- **Pattern A (PA2).** The functional is bounded below and grows at infinity of the admissible class.
  $\longleftrightarrow$ *UCU hypothesis (K).*
- **Pattern A (PA3).** A unique stable saturation exists and is attained.
  $\longleftrightarrow$ *UCU conclusion.*

Remark 9.2 (UCU vs. DS1–DS3). The dissipative selection principle DS1–DS3 of [3] is the *dynamical* partner of UCU: UCU identifies the unique minimiser; DS1–DS3 identifies the unique trajectory to it. In all five instances of Table 1 a time-dependent or flow-parametrised version of J admits the DS1–DS3 structure, but the minimiser itself is already fixed by UCU alone. DS1–DS3 are therefore logically downstream of UCU.

9.4 UCU and the trinity

With UCU in place, the trinity structure of this paper reads:

$$\underbrace{\text{UCU}}_{\text{uniqueness of saturation}} \quad + \quad \underbrace{\text{SGE}}_{\text{HP-status classification}} \quad + \quad \underbrace{\text{Weil-form transversality}}_{\text{branch-independent positivity tool}}$$

UCU provides the *why is there a unique answer* layer; SGE (Section 11) provides the *which branch does the family belong to* layer; the Weil-form architecture (§10) provides the *how to compute positivity in each branch* layer. The three layers are independent: UCU does not predict YES or NO on the HP spectrum; SGE does not predict spectral gaps; Weil-form transversality does not supply uniqueness by itself. They compose to give the full programme.

Remark 9.3 (Scope of UCU). UCU is *necessary but not sufficient* for Hilbert–Pólya realisation. A unique minimiser in the sense of Lemma 9.1 does not automatically provide a Hermitian operator whose spectrum encodes the zeros of Z_X; it provides a ground state. The passage from ground state to spectrum requires the SGE classification (to identify whether one has a group or only a semigroup of transfer operators) and the Weil-form architecture (to recover the explicit formula positivity). UCU is the common denominator, not the proof of Hilbert–Pólya in any single instance.

10 Weil-form methods as branch-transversal complement

We now argue that the Weil quadratic form Q_X of Section 2.3 applies uniformly in all three branch regimes.

10.1 Form-domain anchoring

For each zeta-type family X, the Weil quadratic form Q_X is anchored by the quadruple

$$(Q_X,\ \mathrm{Dom}(Q_X),\ P_X,\ \mathrm{Dom}(Q_X)^{\pm}), \tag{11}$$

where $\mathrm{Dom}(Q_X) \subset L^2(\mathbb{R})$ is a closed subspace on which Q_X is well-defined and lower-bounded (the form domain), P_X is the parity operator $f(t) \mapsto f(-t)$, and $\mathrm{Dom}(Q_X)^{\pm} = \mathrm{Dom}(Q_X) \cap L^2_{\mathrm{even/odd}}$ are the parity-invariant subspaces. For the arithmetic instances of the present paper (Sections 4–6) the form domain is taken as the Friedrichs closure of $C_c^{\infty}([-\log\lambda, \log\lambda])$ with respect to the graph norm of Q_X, and the cosine/sine bases of Section 2.3 realise a density system in $\mathrm{Dom}(Q_X)^{\pm}$. The parity-preservation $P_X\,\mathrm{Dom}(Q_X) = \mathrm{Dom}(Q_X)$ is a consequence of the parity-invariance of the archimedean kernel $e^{u/2}/|2\sinh u|$ (the absolute value is taken; the Weil formula uses the principal-value convention at $u = 0$, cf. [41]) and of the symmetric prime-shift operator family $S_{m\log p} + S_{-m\log p}$ entering the Weil form.

Remark 10.1 (Scope of this anchoring). The explicit form-domain lemma underlying the Riemann instance (restricting Q_ζ to an $H^{1/2}$-closure on which the archimedean kernel is form-bounded, and verifying parity-preservation on the full form domain) is stated as a minor-revision item in [28] (K5). The branch-transversal content of the present paper is independent of that refinement: the Shift Parity Lemma and the Direct Frontier Dominance proof operate on finite-dimensional Galerkin blocks, and their transfer to the continuous operator uses Cauchy-interlacing plus Schur-complement bounds (see Section 2.4, $C^{(2)}$ component), not form-domain regularity per se.

10.2 Positivity at HP-BL-YES

For $X = Z_\Gamma$ (Selberg, Theorem 5.1), the positivity of Q_{Z_Γ} on the relevant test-function class is a consequence of the self-adjointness of $\Delta_{\mathcal{F}}$; the Weil-form positivity can be derived from the spectral decomposition of the Laplacian. This is a consistency check: both architectures give the same answer.

10.3 Positivity at HP-BL-NO

For $X = \zeta$ (Riemann, Theorem 8.1 row 1), the positivity of Q_ζ is established unconditionally in [6] via the Direct Frontier Dominance argument (Shift Parity Lemma + Mertens theorem + CAP-certified base case). The proof uses no Hilbert-Pólya input. This is the strict branch-transversality content.

10.4 Positivity at HP-BL-OPEN

For $X = G_{\text{prime}}$, the Weil-form positivity question splits into two steps. First, the explicit formula $\mathcal{E}_{\text{prime}}$ for Prime-Hub must coincide with the Riemann–Weil explicit formula (5) on the prime/geodesic side, since requirements (OP5.1)–(OP5.3) of Section 6 prescribe that the Fredholm determinant of $\mathcal{L}_s$ equals $1/\zeta(s)$. Consequently $Q_{G_{\text{prime}}} = Q_\zeta$ as a quadratic form on the test-function class, and its positivity reduces to that of Riemann (the HP-BL-NO regime). Second, *provided* an explicit Prime-Hub construction is eventually produced, the Weil-form positivity is automatic by the branch-transversal argument of [6] on the common prime side. This is the strongest possible expression of branch-transversality: the positivity of the quadratic form is invariant under the HP-BL-status transition.

Adjacent evidence for a related HP-BL-OPEN family: Session-8 numerical work on Dirichlet L-functions [9] shows that attempts to build H_χ from phenomenological kernels (Gaussian or Hadamard-weighted pole) fail systematically, while Weil-form positivity remains accessible. This is a second empirical instance where HP-BL-status is effectively NO or OPEN and Weil-form methods continue to apply.

10.5 Summary

The Weil quadratic form is branch-transversal: its existence and positivity-content does not depend on the HP-BL status of the family.

11 The Semigroup-Group Equivalence (SGE)

11.1 A unifying conjecture

The Boundary Theorem 8.1 exhibits three qualitatively distinct values $\{\text{YES}, \text{NO}, \text{OPEN}\}$ of HP-BL on three canonical instances. A natural question is whether this trichotomy is *fundamental* or merely a surface feature of a deeper single criterion. We formulate the following unifying conjecture.

Let $(Z_X, \mathcal{T}_X, \mathcal{E}_X)$ be a zeta-type family in the sense of Definition 2.1. The transfer family $\mathcal{T}_X$ is typically indexed by a discrete set I_X of "prime objects" (rational primes, closed geodesics, primitive cycles), each carrying a canonical length $\ell(\iota)$. The set I_X carries a natural composition operation (multiplication of prime powers, concatenation of geodesics, concatenation of paths), yielding a *transfer algebra* $(I_X, \cdot)$ which is either a *semigroup* (no identity or no inverses) or a *group*.

Conjecture 11.1 (Semigroup-Group Equivalence, SGE)**.** *For every zeta-type family X,*

$$\text{HP-BL}(X) = \text{YES} \quad \Longleftrightarrow \quad (I_X, \cdot) \textit{ is a (locally compact) group.}$$

The backward direction (group ⇒ YES) is classical: a group acting naturally on X produces a Casimir operator that satisfies (HP1)–(HP3). The forward direction (semigroup ⇒ NO) is the substantive content of Conjecture 11.1 and is consistent with every instance tested to date:

Family	Transfer algebra $(I_X, \cdot)$	Predicted HP-BL
Riemann $\zeta(s)$	$\{p^m\}$ (commutative semigroup)	NO (Thm. 8.1)
Dirichlet $L(s,\chi)$	$\{p^m : p \nmid q\}$ (sub-semigroup)	NO (consistent with [9])
Dedekind $\zeta_K(s)$	prime ideals $\{\mathfrak{p}\}$ (ideal monoid)	NO (open testcase)
Selberg $Z_\Gamma(s)$	geodesics $\{\gamma\} \subset \Gamma$	YES (Thm. 5.1)
Ihara $\zeta_G(u)$	closed paths in undirected G	YES (Bass-Hashimoto)
Automorphic $L(s,\pi)$	Hecke algebra on $\Gamma\backslash G$	YES (expected)
Prime-Hub G_{prime}	primitive cycles (free semigroup)	NO (consistent with OP5)

If Conjecture 11.1 holds, the apparent trichotomy of Theorem 8.1 collapses to a single algebraic criterion. The OPEN status of Prime-Hub, in this reading, is not a third fundamental mode but a state of partial knowledge: the explicit construction of $(I_{\text{prime}}, \cdot)$ would deterministically decide between YES and NO via Conjecture 11.1. A separate working note [10] develops the precise formulation and collects falsification criteria.

11.2 First empirical test of Conjecture 11.1: Dedekind zeta of $\mathbb{Q}(\sqrt{-5})$

As a direct probe of Conjecture 11.1 outside the three canonical instances of Theorem 8.1, we performed the following test [11]. For the imaginary quadratic field $K = \mathbb{Q}(\sqrt{-5})$ (discriminant -20, class number 2), the Dedekind zeta ζ_K has a transfer algebra indexed by prime ideals $\mathfrak{p}$ of $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$, which is a semigroup under ideal multiplication. Conjecture 11.1 therefore predicts HP-BL(ζ_K) = NO.

To test this, we enumerated the 92 prime ideals of K with norm at most 500, classified into 2 ramified (above rational primes $2, 5$), 86 split (two ideals for each rational $p \equiv 1, 3, 7, 9 \pmod{20}$), and 4 inert (norm p^2 for $p \equiv 11, 13, 17, 19 \pmod{20}$). The corresponding normalised shifts $r_\mathfrak{p} = 2 \log N\mathfrak{p}/L$ with $L = 10$ yielded 49 distinct values in $(0, 2)$—fewer than the 95 obtained for the rational primes of $\mathbb{Q}$ up to the same norm bound, owing to the split-prime doubling collapsing pairs of ideals to a single shift value. For each shift we constructed the NE-B matrix $D_N(r)$ of Definition 4.3 via numerical quadrature, and computed the symmetric common centraliser dimension by solving the linear system $[M, D_N(r_\mathfrak{p})] = 0$ on the $N(N+1)/2$ symmetric degrees of freedom.

N	$\mathbb{Q}$ (control)	$\mathbb{Q}(\sqrt{-5})$	random shifts
5	$\dim(Z) = 1$	$\dim(Z) = 1$	$\dim(Z) = 1$
7	$\dim(Z) = 1$	$\dim(Z) = 1$	$\dim(Z) = 1$
10	$\dim(Z) = 1$	$\dim(Z) = 1$	$\dim(Z) = 1$

All three columns agree: only the scalar centraliser survives at every tested $N \in \{5, 7, 10\}$. This constitutes the first empirical evidence for Conjecture 11.1 outside the three canonical instances of Theorem 8.1: despite the qualitatively distinct prime decomposition pattern of K and the roughly two-fold lower shift density relative to $\mathbb{Q}$, the Dedekind transfer algebra of $\mathbb{Q}(\sqrt{-5})$ admits the same NE-B behaviour as the rational prime-shift family. The observation supports the algebraic (rather than density-based) nature of NE-B and, transitively, of the HP-BL = NO prediction.

11.3 Complementary empirical test of Conjecture 11.1: Ihara zeta of the Petersen graph

On the HP-BL = YES side of the SGE conjecture, a direct probe is provided by the Ihara zeta function $\zeta_G(u)$ of an undirected finite graph G. The transfer algebra in this case is the path groupoid modulo backtracking, whose associated structure is the fundamental group $\pi_1(G)$—a free group F_r with $r = m - n + 1$ generators. Since F_r is a group (not merely a semigroup), Conjecture 11.1 predicts HP-BL(ζ_G) = YES for every connected undirected G.

We tested this prediction on the Petersen graph (3-regular, $n = 10$, $m = 15$, a Ramanujan graph with $\mathrm{Aut}(G) = S_5$). Three independent verifications were carried out [12]:

(V1) **Bass-Hashimoto identity.** The classical identity $\det(I - uB) = (1 - u^2)^{m-n} \det(I - Au + Qu^2)$, with B the Hashimoto non-backtracking operator on directed edges and $Q = D - I$, was verified at the test point $u = 0.3 + 0.1i$ to 15 significant digits.

(V2) **Ramanujan RH-analog.** 18 of the 19 non-trivial zeros of $\zeta_G(u)^{-1}$ lie exactly on the critical circle $|u| = 1/\sqrt{q-1} = 1/\sqrt{2}$. The single exception at $|u| = 1/2$ corresponds to the trivial adjacency eigenvalue $\lambda_A = 3$.

(V3) **HP-BL = YES certificate.** The graph Laplacian $\Delta_G = D - A$ is self-adjoint with spectrum $\{0, 2^{(5)}, 5^{(4)}\}$. Its eigenvalues relate to the Ihara zeros via the quadratic pencil $2u^2 - \lambda_A u + 1 = 0$. Every graph automorphism $\sigma \in S_5$ induces a permutation matrix P_σ that commutes with both Δ_G and the edge-lift P_σ^{edge} commutes with B. The centraliser of B in $\mathrm{End}(\mathbb{C}^{2m})$ therefore contains the S_5-representation algebra, whose non-scalar elements explicitly exhibit HP-BL$(\zeta_{\mathrm{Petersen}})$ = YES.

In combination with the Dedekind test of Section 11.2, Conjecture 11.1 has now been empirically verified on *both* sides of the semigroup-group dichotomy: the semigroup side ($\mathbb{Q}(\sqrt{-5})$, $\dim(Z) = 1$) at $N \leq 10$, and the group side (Petersen, non-scalar commuting operators constructed explicitly). Five qualitatively distinct zeta-type families now support Conjecture 11.1 consistently (three canonical instances plus these two new tests), and no counter-example has been produced. Elevation to a formally proved theorem would require a structural argument (rather than further case studies) and is identified as the primary next step.

Remark 11.2 (Discriminating control experiment). A natural worry about the Dedekind test of Section 11.2 is that $\dim(Z) = 1$ might be the *generic* centraliser dimension of arbitrary matrix families rather than a structural signature of SGE-NO. A dedicated control experiment [27] addresses this by running the same centraliser-dimension routine on explicit SGE-YES families (cyclic group $\mathbb{Z}/N$ regular representation; elementary abelian $(\mathbb{Z}/2)^k$) alongside a random-symmetric baseline. The outcome is unambiguous:

- $\mathbb{Z}/N$: $\dim(Z_{\mathrm{sym}}) = 7, 10, 14, 17, 22$ at $N = 5, 7, 10, 12, 15$ (grows linearly with N).
- $(\mathbb{Z}/2)^k$: $\dim(Z_{\mathrm{full}}) = 2^k$ exactly (equal to the group order) for $k = 2, 3, 4$.
- Random symmetric baseline: $\dim(Z_{\mathrm{sym}}) = 1$ for all tested N.

The test apparatus therefore discriminates cleanly between SGE-YES (centraliser dimension scales with N) and SGE-NO / null (dimension one). The Dedekind result $\dim(Z) = 1$ is thus structurally informative, not a generic artefact. This closes a documentation gap identified in the 7-phase review [28] of the present manuscript.

Connes' adelic programme [23, 24] fits this picture naturally: by replacing the semigroup $\{p^m\}$ with the idele class group $\mathbb{A}_K/K^\times$, it *transports* the Riemann family from the semigroup column into the group column of the table. On the SGE reading, Connes' programme would upgrade HP-BL(ζ) from NO (in the classical prime-shift transfer algebra) to YES (in the adelic transfer algebra), without contradicting Theorem 8.1.

12 The Blockchain Reading: a speculative bridge

This section is marked as a ***speculative bridge*** *in the four-class rigor hierarchy of Section 13.2: it transports the mathematical content of the trinity into an informatic register through a correspondence with the Bitcoin blockchain protocol. No mathematical assertion of this section is new; the contribution is a consolidated reading that makes the architecture of the trinity accessible to readers familiar with cryptographic protocols.*

12.1 Two algebraic worlds in one protocol

The Bitcoin protocol simultaneously employs two algebraic structures which are, from the perspective of Conjecture 11.1, of opposite type:

1. The **hash function** (SHA-256) generates an image set with a *semigroup* structure under concatenation of data blocks. The hash is deliberately one-way—it has no natural inverse operation—so the indexing set of a hash-chain is a semigroup, not a group.

2. The **digital signature** (ECDSA on the elliptic curve secp256k1) operates on an *abelian group*: the $E(\mathbb{F}_p)$-points form a group under chord-and-tangent addition, with explicit inverses and a well-defined Casimir-like invariant (the curve equation $y^2 = x^3 + 7$).

In the language of Section 11, SHA-256 lives on the semigroup side of the SGE dichotomy, whereas ECDSA lives on the group side. The Bitcoin protocol is thus a *physical realisation* of the algebraic dichotomy that Conjecture 11.1 identifies at the level of zeta-type families, with the hash side mirroring the Riemann prime-power semigroup $\{p^m\}$ and the signature side mirroring the Selberg geodesic group structure $\Gamma \subset \mathrm{PSL}_2(\mathbb{R})$.

12.2 The correspondence table

Tree-theoretic structure	Bitcoin-protocol analog
Prime-power semigroup $I_\zeta = \{p^m : p \text{ prime}, m \geq 1\}$	SHA-256 hash space: semigroup under data concatenation, no natural inverse, provides unique digests (analogous to unique prime factorisation).
Hilbert-Pólya group $I_{Z_\Gamma} = \Gamma \subset \mathrm{PSL}_2(\mathbb{R})$	ECDSA signature group: $E(\mathbb{F}_p)$-points on secp256k1 with chord-and-tangent addition, inverses explicit, Casimir-like curve invariant $y^2 = x^3 + 7$.
Weil quadratic form Q_X (branch-transversal architecture; provides gap positivity regardless of HP-BL-status)	Nakamoto consensus / Proof-of-Work: a protocol-level function that validates the chain uniformly across hash-side and signature-side participants, independently of which algebraic structure an individual transaction uses.
UCU: strict convexity of the governing functional yields a unique minimiser (Section 9)	Longest-chain rule: strictly increasing difficulty along the heaviest chain yields a unique consensus head; cumulative proof-of-work is the strictly convex "score" that selects it.
Theorem 4.3: no universal commuting operator for the prime-shift family at tested Galerkin truncations	Decentralisation: no single Bitcoin node holds a universal validating key; the collective (primes / miners) validates by independent contribution, not by a shared secret.
Riemann hypothesis: all non-trivial zeros on the critical line	Chain consistency: all accepted blocks lie on the canonical (longest, Proof-of-Work-heaviest) chain.
Three-component decomposition $\mathrm{RH}_X = \mathrm{GBC}_X + C^{(2)}_X + \mathrm{Hurwitz}_X$ [6]	Three-layer protocol stack: transaction layer (GBC-analog, spectral content of the operator), consensus layer ($C^{(2)}$-analog, Connes approximation), historical persistence (Hurwitz-analog, reality of zeros preserves the chain past re-orgs).
Conjecture 11.1: HP-BL is a binary algebraic invariant	Protocol hybridism: Bitcoin's functioning requires the cooperation of one semigroup-side component (hashes) and one group-side component (signatures). Each is individually insufficient; their branch-transversal coupling (the protocol itself) is the analog of the Weil quadratic form.

12.3 Reading the trinity cryptographically

The three meta-principles of Section 1.2 receive a compact cryptographic reading:

- **UCU as Proof-of-Uniqueness.** The strict convexity of the governing functional is the protocol-level analog of Bitcoin's longest-chain rule. In both cases, uniqueness of the selected state is guaranteed by a monotone scoring function (difficulty-accumulated work for Bitcoin; the Weil functional for the zeta-family). Non-uniqueness would result in a fork; the convexity condition prevents forks in both domains.

- **SGE as protocol hybridism.** Conjecture 11.1 asserts that each zeta-family lies definitively on one side of the algebraic dichotomy. Bitcoin, by contrast, is a single protocol that employs both sides: its hash layer is HP-BL = NO-structured, its signature layer is HP-BL = YES-structured. In this reading, Bitcoin is the prototype of a *hybrid protocol*—an object whose overall architecture is impossible for a single zeta-family but natural when the semigroup and group sides are coupled via a shared branch-transversal mechanism.
- **Weil-form as branch-transversal consensus.** The Weil quadratic form Q_X functions uniformly for HP-BL = YES and HP-BL = NO families. Nakamoto consensus similarly functions uniformly for hash-side operations and signature-side operations. In both cases, the branch-transversal layer is what allows the two algebraic worlds to cooperate meaningfully.

12.4 Why the correspondence is more than metaphor

Three observations elevate the reading above mere analogy:

1. *The dichotomy is precise.* Bitcoin's hash/signature architecture is a structural necessity, not a design choice: any cryptographic consensus system provably needs both a one-way digesting function (semigroup) and an invertible authentication function (group). This is exactly the content of Conjecture 11.1's binary classification.
2. *Consensus is the algebra.* Nakamoto's insight was that proof-of-work consensus can function as a branch-transversal layer uniting the two sides. The Weil quadratic form plays the same role in the zeta-family tree: it is the common arithmetic on which both HP-BL-regimes operate.
3. *The uniqueness theorem is shared.* Bitcoin's longest-chain rule is *mathematically* a strictly convex selection principle; the Weil functional's strict convexity is the same principle in the continuous setting. Both are instances of a general convexity-implies-uniqueness argument (UCU).

The correspondence, accordingly, is not a decorative metaphor but a structural echo. The same trinity—UCU, SGE, Weil-form transversality—describes both the classification of zeta-type programmes and the architecture of the Bitcoin blockchain.

12.5 Rigor disclaimer and further reading

The present section is a *speculative bridge.* It introduces no new mathematical theorem and no new cryptographic result; the correspondences above are conceptual pairings, not identities. What the section does provide is a consolidated register in which the trinity's structural content can be communicated to readers from cryptography, computer science, and decentralised-systems research. For a fuller informal development of the blockchain analogy in the broader tree-of-zeta-families context (including related motifs on primes as holographic boundary data and on time as information accumulation), see the companion notes [2], Appendix I8–I10, and the supplementary working notes referenced there.

13 Discussion and outlook

13.1 Consequences for Dirichlet L-functions

Session-8 work on Dirichlet L-functions [9] established that Paley-Wiener transfer via phenomenological kernels (Gaussian or Hadamard-weighted pole) does not capture the character-specific gap signatures. The boundary theorem (Theorem 8.1) interprets this result: for low-conductor real Dirichlet characters, no natural HP-BL operator is constructed by simple perturbation of the Riemann case, and the observed character-specific gaps must arise from the branch-transversal Weil-form architecture, whose leading-order reductions have been shown to fail. A dedicated Dirichlet-L atlas paper [8] documents this failure in detail.

Under Conjecture 11.1, this outcome is structurally predicted: the Dirichlet transfer algebra is a semigroup, so HP-BL($L(\cdot, \chi)$) = NO and no natural HP operator can exist. The Dirichlet atlas is therefore not a failure of the programme but a confirmation of its architectural boundaries.

A post-Zookeeper refinement is now part of the CORE status. The Spectral Zookeeper [14] supplies an independent kernel package for the Riemann cell: rank-one decomposition, Cauchy interlacing, three-lemma closure, and spectral microclusters. This does not change the boundary theorem; it refines the UBA layer. The Riemann cell now carries two orthogonal kernel packages: v2.1 even dominance as the published C2-sensitive layer, and the Zookeeper three-lemma package as the theorem-level closure of the Riemann $C^{(2)}$/MS2 step in the CCM/Fourier branch. For the Dirichlet cell the natural back-transfer is Route D: a χ-twisted CCM operator with a conductor rank-one term and a χ-weighted arithmetic perturbation. Thus the Dirichlet atlas should be read as the first concrete UBA stress test of this transfer, not merely as a negative Paley-Wiener numerical result.

13.2 Proposed rigor hierarchy

Following the four-class hierarchy advocated in the tree meta-analysis (theorem-level / conditional / diagnostic / speculative), each component of the branch-locality programme can be classified as follows:

Component	Class	Source
Definition 2.3	theorem-level	this paper
Theorem 4.1	theorem-level	[6]
Theorem 4.3 ($N \leq 15$)	theorem-level	[6]
Conjecture 4.4	conditional	[6]
Theorem 5.1	theorem-level	[40]
Props. 6.1–6.5	diagnostic	[7]
Theorem 8.1	theorem-level*	synthesis
Conjecture 11.1 (SGE)	conditional (empirical)	this paper, [10]
§12 (Blockchain reading)	speculative bridge	this paper

* conditional on Conjecture 4.4.

13.3 Future directions

1. **Formal proof of Conjecture 4.4.** An SVD-density argument for $N = \infty$ is the single most valuable next step: it would upgrade Theorem 8.1 from conditional to unconditional.

2. **Test Conjecture 11.1 on Dedekind zeta.** The NE-B argument of [6] relies on structural properties of the prime shift operators that extend to prime-ideal shift operators over a number field K. A direct SVD-density test would either confirm or refute the SGE prediction HP-BL(ζ_K) = NO.

3. **Ihara zeta and Yang-Lee roots.** Ihara's case is the graph-theoretic analogue of Selberg's and is expected by Conjecture 11.1 to satisfy HP-BL = YES, with the Bass-Hashimoto matrix serving as H_X. Yang-Lee zeros, by contrast, are partition-function roots parameterised by temperature and should on present evidence sit in the flow-zeta class of Section 3.4, as a statistical-mechanical relative of CRM.

4. **Population of the zoo.** The six zoo entries exhibited so far (Riemann, Selberg, Prime-Hub, CRM, Dedekind $\mathbb{Q}(\sqrt{-5})$, Ihara-Petersen) plus the Dirichlet atlas companion [8] are illustrative, not exhaustive. A scan of the surrounding literature and the companion tree documents [17] identifies eight further families that fit the three-class taxonomy cleanly and sharpen the SGE dichotomy:

   - **Automorphic $L(\pi, s)$ for** GL_n (Langlands programme): geometric-compact, **SGE-YES**. Transfer algebra is the Hecke algebra; Casimir of the ambient reductive group supplies the Hilbert-Pólya operator. The canonical large-family residence of the YES-side.
   - **Hecke $L(s, \lambda)$ over number fields**: arithmetic, **SGE-NO**. Sister species to Dirichlet and Dedekind—prime-ideal semigroup plus character twist.
   - **Artin L-functions**: geometric-compact, **SGE-YES**. Transfer algebra is a finite Galois group; prototype of the Galois-driven YES-side.
   - **Ruelle-Anosov zeta** $\zeta_\varphi(s)$ for hyperbolic flows: flow-noncompact, **SGE-YES**. Direct generalisation of CRM beyond the $\mathrm{PSL}_2(\mathbb{R})$-boost case; see [16].
   - **Selberg on the non-compact quotient** $\mathrm{PSL}_2(\mathbb{R})/\mathrm{SO}(2) = \mathbb{H}$: flow-noncompact, **SGE-YES**. Continuous spectrum plus Eisenstein series. Algebraically almost identical to CRM, dynamically distinct.
   - **Black-hole quasinormal-mode spectra** ([35]): flow-noncompact, **SGE-YES**. Scattering resonances on asymptotically de Sitter backgrounds; the Wheeler–DeWitt wavefunctions of 5d BKL dynamics exhibit automorphic-L and complex-primon-gas structures linking zeta-like spectra to quantum-gravity regimes. The first flow-zeta with realistic hopes of experimental access via gravitational-wave ringdown observations (LIGO/Virgo/Einstein Telescope).
   - **Yang-Lee zeros**:
     ferromagnetic partition functions; conjecturally flow-noncompact, **SGE-YES**. Temperature as continuous parameter; the Lee-Yang circle theorem plays the role of an RH-analog. Statistical-mechanical cousin of CRM.
   - **Bost-Connes system** (1995): a quantum-statistical realisation of $\zeta(s)$ as partition function with spontaneous symmetry breaking at $\beta = 1$. Sits on the bridge between the arithmetic and flow classes and connects the zoo directly to Connes' adelic framework [23, 24].

   Each of these deserves a dedicated FST-Mathematics supplement in due course. A second tier of twelve further constructions (Epstein, Witten-zeta, multiple-zeta-values, Hurwitz, Lerch, Arakelov, motivic L, chiral-zeta, RG-zetas, Gromov-Witten, Reidemeister torsion, QFT spectral zetas) and a separate column of biological Pattern-A-derived zetas (Kauffman-attractor, protein-folding spectral, replicator Jacobian) are catalogued in the companion tree document [17].

5. **Biological Pattern-A instances as bridges.** The FST-Biology supplement programme [18] develops Pattern-A in molecular, chemical, and behavioural domains (protein folding as a Nash equilibrium, gene regulatory network attractors, immune-system coevolution). These are primarily residents of the physical side of the programme, but generate zeta-like counting functions of their own (Kauffman-attractor zetas for Boolean gene networks, protein-folding spectral zetas, replicator Jacobian spectra) that could in principle populate dedicated enclosures in the zoo. Their primary classification is on the physical side of FST; their mathematical structure is listed in the companion candidate document.

6. **Adelic upgrade: Riemann revisited.** Connes' adelic framework [23, 24] replaces the classical prime-power semigroup $\{p^m\}$ with the idele class group $\mathbb{A}_K/K^\times$, which *is* a group. If that

programme closes, Riemann migrates from the SGE-NO column (classical prime-shift transfer algebra) to the SGE-YES column (adelic transfer algebra), without contradicting Theorem 8.1—the SGE conjecture describes precisely what a successful Hilbert-Pólya programme must achieve.

7. **Explicit Prime-Hub construction.** Any future construction must explicitly navigate Propositions 6.1–6.5. Under Conjecture 11.1, the construction of G_{prime} succeeds in producing a Hilbert-Pólya operator if and only if the resulting cycle-concatenation algebra is a group.

8. **Independent replication and a reviewer-verifiability index.** The trilogy [6, 7] currently comprises ~82 PDF pages, 76 theorem objects, an eight-step reduction chain, 33 CAP certificates, and twenty Python scripts. The critique review [28], while finding no fatal gap, identifies documentation-ergonomic weaknesses (K6): absence of a central symbol glossary, no dependency graph A1–A8, no unified `reproduce.sh` pipeline, no "reviewer fast path" note. We propose a *Reviewer Kit* companion Zenodo record containing (a) a symbol glossary, (b) a dependency DAG linking lemmas to scripts to A-steps, (c) a `scripts/REPRODUCE.md` with one-liner invocations and expected runtimes, (d) a Lean4 skeleton for the Shift Parity Lemma—its purely algebraic structure ($N \geq 3$, closed-form determinant and trace) makes it the natural first target for serious formalisation, and (e) a two-page "reviewer fast path" stating how each A-step can be independently verified in a bounded time budget. An accompanying *reviewer-verifiability index* per A-step (analytic / CAP / hybrid, estimated person-hours for independent verification) would make the trilogy's replication cost explicit. Crucially, this programme requires no change to the trilogy itself; it is an additive companion that addresses K6 without touching the proof.

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References

[1] L. Geiger, *The Riemann Hypothesis via the v2.0 Method, Part I: Foundations and Obstructions*, Preprint (2026), Zenodo DOI 10.5281/zenodo.19546593.

[2] L. Geiger, *A Meta Galerkin Bridge: Diagnostic Decomposition of RH-Type Programmes*, Working draft (2026), `paper/META_GALERKIN_BRIDGE.tex` in the archived predecessor tree. Appendix I collects the ten essay-level motivations referenced here, including the blockchain-themed items I8–I10.

[3] L. Geiger, *Functional Stability Theory — Physical Instantiation via the Renormalized Free Energy Principle*, Preprint, companion paper (2026), repo research-line/functional-stability-theory (formerly published as "The Renormalized Free Energy Framework" v1.7, to be re-issued as v1.8 under the unified FST terminology).

[4] L. Geiger, *FST-Physics: Yang–Mills Mass Gap via Pattern A*, Preprint (2025/2026), Zenodo DOI 10.5281/zenodo.19087433.

[5] L. Geiger, *FST-Physics: Kolmogorov K41 Spectrum via Pattern A*, Preprint (2025/2026), Zenodo DOI 10.5281/zenodo.19056813.

[6] L. Geiger, *The Riemann Hypothesis via the v2.0 Method, Part II: Even Dominance (v2.1 Unconditional)*, Preprint (2026), Zenodo DOI 10.5281/zenodo.19546593.

[7] L. Geiger, *The Riemann Hypothesis via the v2.0 Method, Part III: Conclusio and Open Problems*, Preprint (2026), Zenodo DOI 10.5281/zenodo.19546593.

[8] L. Geiger, *A Weil-Kernel Numerical Atlas for Dirichlet L-Function Sector Gaps: Galerkin Diagnostics and the Boundaries of Leading-Order Theory*, Draft (2026), `CORE/zoo-mapping/paper/DIRICHLET_CHARACTER_ATLAS_v1_en.tex`.

[9] L. Geiger, *Paley-Wiener Transfer for Dirichlet L-functions: First-Order Perturbation and Numerical Validation*, Working notes (2026), `CORE/zoo-mapping/PALEY_WIENER_DIRICHLET_TRANSFER.md`.

[10] L. Geiger, *The Semigroup-Group Equivalence: A Unifying Conjecture for Branch-Locality*, Working notes (2026), `paper_NE_B_BOUNDARY/SEMIGROUP_GROUP_EQUIVALENCE.md`.

[11] L. Geiger, *Dedekind NE-B Analog Test for $\mathbb{Q}(\sqrt{-5})$: First Empirical Probe of the Semigroup-Group Equivalence*, Computational notes (2026), `paper_NE_B_BOUNDARY/_results/DEDEKIND_NE_B_TEST.md`.

[12] L. Geiger, *Ihara-Zeta SGE YES-side Test for the Petersen Graph: Bass-Hashimoto, Ramanujan, and the Laplacian as HP-Operator*, Computational notes (2026), `paper_NE_B_BOUNDARY/_results/IHARA_PETERSEN_SGE.md`.

[13] L. Geiger, *CRM as Flow-Zeta: Reformulation of the Cooperative Renormalization Model of Dark Energy as a Ruelle-Dyatlov-Zworski Zeta on the Hyperbolic Plane*, Working note (2026), `04_FST_COSMOLOGY/CRM_AS_FLOW_ZETA.md`.

[14] L. Geiger, *The Spectral Zookeeper: The Riemann Hypothesis via Spectral Microcluster Closure in the Connes–Consani–Moscovici Framework*, Preprint (2026), Zenodo DOI 10.5281/zenodo.19673127.

[15] L. Geiger, *Cooperative Renormalization Model of Dark Energy V: The Saturation Theorem*, Preprint (2026), Zenodo concept DOI 10.5281/zenodo.18728936.

[16] S. Dyatlov and M. Zworski, *Mathematical Theory of Scattering Resonances*, Graduate Studies in Mathematics **200**, American Mathematical Society, Providence, RI (2019), DOI 10.1090/gsm/200.

[17] L. Geiger, *Zeta Zoo Candidates — Master List*, Companion tree document (2026), `CORE/CANDIDATES_MASTER_LIST.md` in the `.ZETA-ZOO` project tree. Maintains the full catalogue of zeta-type family candidates identified across RH v2.1, EFP-Taxonomy, V2-Physics-Testcases, FST-Biology, and supplementary Ideenanhang sources.

[18] L. Geiger, *Functional Stability Theory — Biological Pattern-A Instances (FST-I Thermodynamic, FST-II Chemical, FST-III Biological Stability)*, Preprint series (2026), `05_FST_BIOLOGY/` in the `.ZETA-ZOO` project tree. Three-paper trilogy on scale-invariant Pattern-A realisations from molecular to behavioural biology.

[19] G. Pólya, letter to A. M. Odlyzko, 3 January 1982, recounting a Göttingen conversation with E. Landau (ca. 1912–1914) about a possible operator-theoretic explanation for the Riemann hypothesis. Preserved by Odlyzko at `https://www-users.cse.umn.edu/~odlyzko/polya/`.

[20] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in: *Analytic Number Theory* (Proc. Sympos. Pure Math. **XXIV**, St. Louis 1972), pp. 181–193, American Mathematical Society, Providence, RI (1973). Earliest published mention of the Hilbert–Pólya suggestion as folklore.

[21] M. V. Berry, *Semiclassical theory of spectral rigidity*, Proc. Roy. Soc. London A **400**, 229–251 (1985), DOI 10.1098/rspa.1985.0078.

[22] M. V. Berry and J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, SIAM Rev. **41** (2), 236–266 (1999), DOI 10.1137/S0036144598347497.

[23] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (New Series) **5** (1), 29–106 (1999), DOI 10.1007/s000290050042.

[24] A. Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv preprint (2026), arXiv:2602.04022 [math.NT], `https://arxiv.org/abs/2602.04022`.

[25] A. Connes, C. Consani, and H. Moscovici, *Zeta Spectral Triples*, arXiv preprint (2025), arXiv:2511.22755 [math.NT], `https://arxiv.org/abs/2511.22755`. Companion paper to [24].

[26] L. Geiger, *Obstruction Classifiers for RH-Type Problems*, Working paper (2026), cf. `_archive/old_obstruction_classifier_wrapper/Obstruction_Classifiers_v1_en.tex` in the archived predecessor tree.

[27] L. Geiger, *SGE Control Experiment — Discriminating Test*, Working note and computational artefact (2026), `CORE/zetazoo/_results/SGE_CONTROL_EXPERIMENT.md`, produced by `CORE/zetazoo/_scripts/sge_control_experiment.py`. Server-certified run (ellmos-services, 2026-04-17) reporting centraliser-dimension scaling for cyclic $\mathbb{Z}/N$, elementary abelian $(\mathbb{Z}/2)^k$ and random-symmetric baselines.

[28] L. Geiger, *RH v2.1 — Systematic Critique Review (K1–K6)*, Working note (2026), `SESSIONS/_reviews/RH_CRITIQUE_REVIEW_2026-04-17.md` in the `.ZETA-ZOO` project tree. Four-agent parallel assessment of the Foundation–Even-Dominance–Conclusio trilogy against six external critiques; verdict: all partially valid, none fatal.

[29] I. Ekeland and R. Témam, *Convex Analysis and Variational Problems*, SIAM Classics in Applied Mathematics **28**, Society for Industrial and Applied Mathematics, Philadelphia (1999), ISBN 978-0-89871-450-0. Standard reference for Mazur's lemma and the strict-convexity-plus-coercivity uniqueness principle used in Lemma 9.1.

[30] J. P. Keating and N. C. Snaith, *Random Matrix Theory and* $\zeta(1/2+it)$, Commun. Math. Phys. **214** (2000), pp. 57–89, DOI 10.1007/s002200000261.

[31] E. Bombieri, *Problems of the Millennium: The Riemann Hypothesis*, Clay Mathematics Institute Official Problem Description (2000), `https://www.claymath.org/wp-content/uploads/2022/05/riemann.pdf`.

[32] J. B. Conrey, *The Riemann Hypothesis*, Notices of the AMS **50** (2003), no. 3, pp. 341–353, `https://www.ams.org/notices/200303/fea-conrey-web.pdf`.

[33] J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. (New Series) **1** (1995), no. 3, pp. 411–457, DOI 10.1007/BF01589495.

[34] R. P. Langlands, *Problems in the theory of automorphic forms*, in: C. T. Taam (ed.), *Lectures in Modern Analysis and Applications III*, Lecture Notes in Mathematics **170**, Springer, Berlin–Heidelberg (1970), pp. 18–61, DOI 10.1007/BFb0079065.

[35] M. De Clerck, S. A. Hartnoll, and M. Yang, *Wheeler–DeWitt wavefunctions for 5d BKL dynamics, automorphic L-functions and complex primon gases*, arXiv preprint (2025), arXiv:2507.08788 [hep-th], https://arxiv.org/abs/2507.08788.

[36] E. Artin, *Zur Theorie der L-Reihen mit allgemeinen Gruppencharakteren*, Abh. Math. Sem. Univ. Hamburg **8** (1931), pp. 292–306, DOI 10.1007/BF02941010.

[37] N. Tschebotareff, *Die Bestimmung der Dichtigkeit einer Menge von Primzahlen, welche zu einer gegebenen Substitutionsklasse gehören*, Math. Ann. **95** (1926), pp. 191–228, DOI 10.1007/BF01206606.

[38] J. W. Cogdell and I. I. Piatetski-Shapiro, *Converse theorems for GL_n*, Publ. Math. Inst. Hautes Études Sci. **79** (1994), pp. 157–214, https://www.numdam.org/item/PMIHES_1994__79__157_0/.

[39] S. Dyatlov and M. Zworski, *Dynamical zeta functions for Anosov flows via microlocal analysis*, Ann. Sci. Éc. Norm. Supér. (4) **49** (2016), no. 3, pp. 543–577, DOI 10.24033/asens.2290, arXiv:1306.4203.

[40] A. Selberg, *Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces with applications to Dirichlet series*, J. Indian Math. Soc. **20**, 47–87 (1956).

[41] A. Weil, *Sur les "formules explicites" de la théorie des nombres premiers*, Meddelanden från Lunds Universitets Matematiska Seminarium, Supplementband tillägnet Marcel Riesz, pp. 252–265, C. W. K. Gleerup, Lund (1952).

[42] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications **53**, AMS, Providence, RI (2004), xi+615 pp., ISBN 0-8218-3633-1.

[43] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd edition, revised by D. R. Heath-Brown, Oxford University Press, Oxford (1986), ISBN 0-19-853369-1.